Eckardt's self-excited homopolar generator

Does the twin-disk machine's “giant resonance” draw energy from spacetime, or from the shaft? — register entry H15, CLAUDE.md §3f

kronspice simulation report · 2026-11-07

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Summary

H. Eckardt (A.I.A.S./UPITEC, 2012) describes a twin-disk homopolar machine that excites its own field and reports an energy surplus at a “giant resonance” where the effective resistance vanishes. This example runs the paper's equations as printed, then the same machine as a kronspice circuit built on the N machine element, and compares their energy books.

- **The resonance is reproduced:** eq. (9) gives ω_{res} = 29.84 rad/s (paper: about 30). It is the self-excitation threshold of a series-wound dynamo.
- **The paper's equations run away:** the current passes 10^{6} A at 0.463 s, and the energy into R and the coil is 1.42×10^{9} J against 10.7 J of drive work.
- **The surplus is the omitted back torque:** the missing 1.423×10^{9} J equals, to 10^{-3}, the work the back torque $\frac{1}{2}r^2(B_m + B_I)\cdot I$ would have taken from the shaft, which eq. (12) leaves out.
- **With the back torque the current stays bounded:** in kronspice the speed settles at ω_{res} with I ≈ 9.949 A and $P = \alpha\cdot\omega_{res}$ = 29.84 W from the drive; the ledger closes to 3×10^{-5} J.
- **The paper's flywheel pulse is the flywheel's energy:** what R receives is what the shaft gave (books close to 10^{-8} J).

What is being tested

Hypothesis (H15). ECE theory (Evans; A.I.A.S. UFT paper 107) holds that homopolar resonances draw energy from spacetime. Eckardt's 2012 paper, “Simulation of resonances in the homopolar generator”, models a self-excited twin-disk machine and reports a “giant resonance” at R_{eff} = 0 with an energy surplus.

Standard prediction. R_{eff} = 0 is the self-excitation threshold of a series-wound dynamo (Siemens 1867; Bullard 1955; Moffatt 1978, ch. 6). Above it the current grows by taking power from the drive. The Lorentz force that gives the disk EMF $\frac{1}{2}\omega r^2 B$ also gives a back torque $\frac{1}{2}r^2B\cdot I$ per disk, so $\omega\cdot$(torque) = EMF·I exactly and the shaft pays for every joule.

The machine and the paper's equations

Two disks of radius r on one shaft are in series with a resistance R and a coil of N turns, length l and core permeability μ_r (paper Fig. 1). The “static” disk sits in a permanent-magnet field B_m; the “dynamic” disk sits in the coil's own field $B_I = \mu_0\mu_r NI/l$, so the loop current I excites it. ω is the shaft speed, J the inertia and α the drive torque.

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L·dI/dt + R·I = ½·ω·r²·(Bm + B_I)          (6)
R_eff = R − ½·ω·r²·μ0·μr·N/l               (8)
ω_res = 2·R·l/(r²·μ0·μr·N)                 (9)
J·dω/dt = α   (constant drive torque)       (12)  as printed
J·dω/dt = α − ½·r²·(Bm + B_I)·I                   with the back torque
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parameter (paper p. 4)	value	parameter	value
R	0.3 Ω	r	10 cm
L	1 mH	μ_r	100
N	800	B_m	0.1 T
l	5 cm	J	0.01 kg·m²

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parameter (paper p. 4)	value	parameter	value
α	1 N·m	ω_{res} from eq. (9)	29.84 rad/s (paper: about 30)

In kronspice the machine is one N element with four windings (static disk path, dynamic disk path, coil, and a magnet equivalent at 1 A). Its speed-matrix entries are K1 = ½r²B_m = 0.0005 V·s and K2 = ½r²$\mu_0\mu_r$N/l = 0.01005 V·s/A; the shaft is a mechanical node with inertia J and the drive torque α. The back torque is part of the element (Kron's speed-voltage tensor), not an added term.

Results

1. The paper's equations as printed (RK4, no back torque)

The speed rises linearly, and once ω passes ω_{res} the current runs away: 0.56 A at 0.30 s, 211 A at 0.40 s, 1.2×10^5 A at 0.45 s, over 10^6 A at 0.463 s. The energy into R and the coil is 1.42×10^9 J against 10.7 J of drive work, all of it spin-up. The missing 1.423×10^9 J equals, to 10^{-3}, the work the omitted back torque ½r²(B_m + B_l)·I would have taken from the shaft. The “energy from spacetime” in these equations is the shaft work they leave out.

2. The same machine in kronspice (back torque inside the element), L = 1 mH, 3 s

t	ω	I	B_l
0.30 s	30.08 rad/s	0.579 A	1.16 T
0.60 s	25.77 rad/s	5.05 A	10.2 T
1.20 s	27.37 rad/s	12.69 A	25.5 T
3.00 s	30.33 rad/s	9.68 A	19.5 T

The current stays bounded and the speed settles at ω_{res}. Energy ledger: drive work 84.75 J = R 80.1 J + coil 0.047 J + spin-up 4.6 J + reference resistors 0.0002 J; it closes to 3×10^{-5} J. Steady state: α = K1·I + K2·I² gives I = 9.949 A and P = α·ω_{res} = 29.84 W = R·I² (29.69 W).

That steady state needs B_l = 20 T in the iron, which saturates near 1.5–2 T, so a real coil limits the excitation well before. With the coil inductance the stated geometry implies (50.5 H if B_l fills the disk area) the run is much slower and the shaft reverses as the field hunts; the books still close.

3. The paper's second run: disk spun to 40 rad/s, then connected (Figs. 6–9)

drive	peak current	speed after 0.5 s	energy into R
α = 0	32.34 A at 0.057 s	19.53 rad/s	6.092 J = flywheel 6.092 J
α = 1 N·m	46.32 A at 0.049 s	25.50 rad/s	19.197 J = flywheel 4.749 J + drive 14.466 J − coil 0.017 J

The pulse is the flywheel's energy discharged through the self-excited dynamo: what R receives is what the shaft gave (books close to 10^{-8} J).

Ledger and validation

- ω_{res} from eq. (9) against the paper's value.
- The energy missing from the paper's books equals the work of the omitted back torque, to 10^{-3}.
- With the back torque the current is bounded and the ledger closes (3×10^{-5} J in run 2; 10^{-8} J in the flywheel runs).
- *main_test.go* covers all of these; *ECKARDT_DUMP=1* prints the comparison series. *export_test.go* writes the driven and flywheel netlists, KiCad schematics and results sheets to *export/*.

Limits of the model

- The coil field is linear ($B_l = \mu_0\mu_r NI/l$); the 20 T steady state is far above iron saturation (1.5–2 T), so a real machine would limit the excitation much earlier.
- The main runs use the paper's L = 1 mH, not the 50.5 H its geometry implies.
- The two ECE-only resonances of UFT 107 are not modelled, because no equations are supplied for them.

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Status

Register status: **untested**. The classical resonance ($R_{eff} = 0$) is the series-dynamo self-excitation threshold and is reproduced. The paper's energy surplus equals, to 10^{-3}, the work of the back torque its eq. (12) omits; with it the current stays bounded (≈ 9.9 A at ω_{res}, 30 W from the drive) and the ledger closes. The two ECE-only resonances are not modelled.

Reproducing the results

```
go run ./examples/eckardt_homopolar
go test ./examples/eckardt_homopolar
python examples/eckardt_homopolar/viz/make_report.py examples/eckardt_homopolar/eckardt_homopolar_report.pdf
```

Sources: H. Eckardt, “Simulation of resonances in the homopolar generator,” A.I.A.S./UPITEC, 13 February 2012 (hypothesis source); E. C. Bullard, Proc. Camb. Phil. Soc. 51, 744 (1955); H. K. Moffatt, Magnetic Field Generation in Electrically Conducting Fluids, Cambridge 1978, ch. 6; G. Kron, Equivalent Circuits of Electric Machinery, Wiley 1951, Epilogue.

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